

Chapter 5: Differentiation

Section: Taylor's Theorem

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

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The Taylor Polynomial

Taylor's Theorem

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Summary

Motivation



From the Mean Value Theorem to Taylor

- The mean value theorem (5.10), rewritten:

$$f(\beta) = \underbrace{f(\alpha)}_{\text{constant approximation}} + \underbrace{f'(x)(\beta - \alpha)}_{\text{exact error term}} \quad \text{for some } x \text{ between } \alpha, \beta$$

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approximate f by a **polynomial of degree $n - 1$** built at α

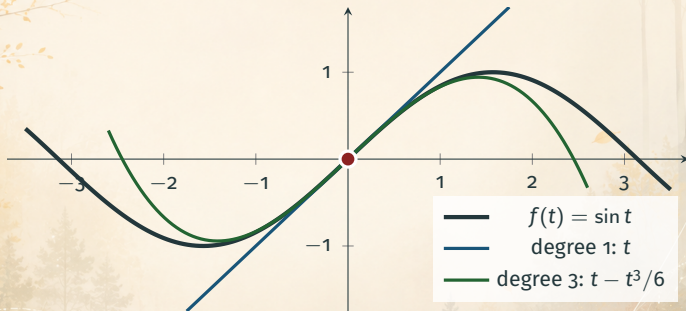
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- With n derivatives available, we can do better:
approximate f by a **polynomial of degree $n - 1$** built at α
- **Taylor's theorem:** the error keeps the same shape — one n th derivative at one unknown point: $\frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$

Taylor Polynomials Hug the Curve



Both polynomials are built **only** from derivatives of f at the red point $\alpha = 0$.

The background is a serene, misty forest landscape. In the foreground, a calm lake reflects the surrounding trees and the hazy sky. The trees are mostly evergreens, with some deciduous trees showing autumn foliage. The sky is a soft, warm yellow-orange, suggesting a sunrise or sunset. Overlaid on the landscape are several abstract geometric elements: thin white lines forming a grid-like pattern, and glowing yellow lines that curve and swirl across the scene, creating a sense of movement and depth. The overall mood is peaceful and ethereal.

The Taylor Polynomial

The Setup for Theorem 5.15

Standing hypotheses

- f is a **real** function on $[a, b]$; n is a positive integer
- $f^{(n-1)}$ is **continuous** on $[a, b]$
- $f^{(n)}(t)$ exists for every $t \in (a, b)$

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Two distinct points

α, β are **distinct** points of $[a, b]$:

α = where we *know* f (center of expansion)

β = where we *want* f (target)

The Taylor Polynomial

The Taylor polynomial of f at α

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k \quad (23)$$

Written out:

$$P(t) = f(\alpha) + f'(\alpha)(t - \alpha) + \frac{f''(\alpha)}{2!}(t - \alpha)^2 + \cdots + \frac{f^{(n-1)}(\alpha)}{(n-1)!}(t - \alpha)^{n-1}$$

Why P Is the Right Polynomial

P copies all available derivatives of f at α

$$P^{(k)}(\alpha) = f^{(k)}(\alpha) \quad \text{for } k = 0, 1, \dots, n-1$$

Reason: differentiating $\frac{(t - \alpha)^j}{j!}$ exactly k times and setting $t = \alpha$ gives 1 if $j = k$, and 0 otherwise. Each coefficient of P is tuned to match one derivative.



Taylor's Theorem

Theorem 5.15: Taylor's Theorem

Theorem 5.15 (Taylor)

Under the standing hypotheses, there exists a point x between α and β such that

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!} (\beta - \alpha)^n. \quad (24)$$

Key idea: the value of f at β = polynomial prediction + an **exact** error term of mean-value type.

The Case $n = 1$: The Mean Value Theorem

Taylor with $n = 1$

The sum (23) has the single term $k = 0$, so $P(t) = f(\alpha)$, and (24) reads

$$f(\beta) = f(\alpha) + f'(\alpha)(\beta - \alpha).$$

This is exactly the **mean value theorem** (5.10).

Taylor's theorem = *the mean value theorem of order n .*

What the Theorem Buys Us: Error Estimates

From an unknown point to a usable bound

If $|f^{(n)}(t)| \leq M$ for all t between α and β , then (24) gives

$$|f(\beta) - P(\beta)| \leq \frac{M}{n!} |\beta - \alpha|^n.$$

- f can be **computed** to any accuracy from finitely many derivatives
- $n!$ grows extremely fast — the bound improves rapidly with n

Proof of Taylor's Theorem

Proof Strategy

Goal: find x between α and β with $f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$.

Approach:

1. Let M be the number that makes $f(\beta) = P(\beta) + M(\beta - \alpha)^n$ true.
Then show $n! M = f^{(n)}(x)$ for some x between α and β
2. Build the auxiliary function $g(t) = f(t) - P(t) - M(t - \alpha)^n$
3. Reduce the goal to: $g^{(n)}(x) = 0$ for some x
4. g vanishes n times over: repeated mean value theorem produces such an x

Technique: auxiliary function + iterated mean value theorem.

Proof of Theorem 5.15 — Step 1: Define M and g

Step 1: Set up

Let M be the (unique) number defined by

$$f(\beta) = P(\beta) + M(\beta - \alpha)^n \quad (25)$$

and put

$$g(t) = f(t) - P(t) - M(t - \alpha)^n \quad (a \leq t \leq b). \quad (26)$$

To show: $n! M = f^{(n)}(x)$ for some x between α and β .

Proof of Theorem 5.15 — Step 2: Reduce to a Zero of $g^{(n)}$

Step 2: Differentiate (26) n times

P has degree $\leq n - 1$, so $P^{(n)} \equiv 0$; $[(t - \alpha)^n]^{(n)} = n!$

$$g^{(n)}(t) = f^{(n)}(t) - n! M \quad (a < t < b). \quad (27)$$

New goal: $g^{(n)}(x) = 0$ for some x between α and β .

Then (27) gives $n! M = f^{(n)}(x)$.

Proof of Theorem 5.15 — Step 3: g Vanishes Many Times

Step 3: Zeros of g and its derivatives

Since $P^{(k)}(\alpha) = f^{(k)}(\alpha)$ for $k = 0, \dots, n-1$:

$$g(\alpha) = g'(\alpha) = \dots = g^{(n-1)}(\alpha) = 0. \quad (28)$$

Moreover, the choice (25) of M gives $g(\beta) = 0$.

g vanishes at **both** α and β , and its first $n-1$ derivatives vanish at α .

Proof of Theorem 5.15 — Step 4: The Cascade

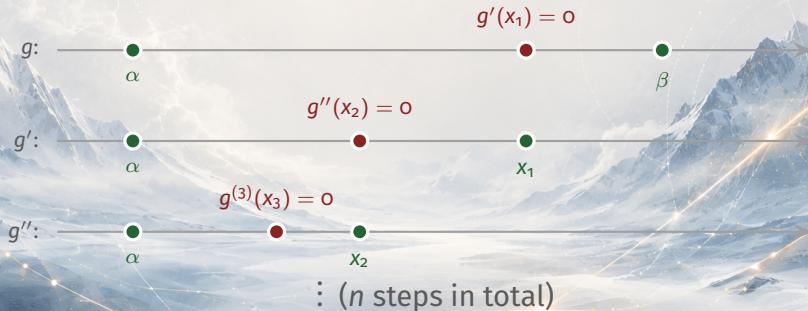
Step 4: First application of the mean value theorem

$g(\alpha) = g(\beta) = 0 \implies g'(x_1) = 0$ for some x_1 between α and β .

Proof of Theorem 5.15 — Step 4: The Cascade

Step 4: Iterate, using (28) at each stage

$g'(\alpha) = g'(x_1) = 0 \implies g''(x_2) = 0$ for some x_2 between α and x_1
... after n steps: $g^{(n)}(x_n) = 0$ for some x_n between α and x_{n-1} .



Proof of Theorem 5.15 — Conclusion

Conclusion

Each x_k lies between α and x_{k-1} , hence between α and β . Take $x = x_n$:

$$g^{(n)}(x) = 0 \xrightarrow{(27)} n!M = f^{(n)}(x) \xrightarrow{(25)}$$

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$$

This proves (24). ■

The background is a soft-focus landscape featuring a calm lake reflecting distant mountains and cherry blossom trees. A complex, golden geometric pattern, consisting of intersecting lines and circles, is overlaid on the scene. The word "Summary" is centered in the upper half of the image, with a horizontal line extending from its base across the width of the slide.

Summary

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Key takeaways from this section:

- **Theorem 5.15 (Taylor):** $f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$ for some x between α and β , where $P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t - \alpha)^k$

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- **Theorem 5.15 (Taylor):** $f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$ for some x between α and β , where $P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t - \alpha)^k$
- **It generalizes the MVT** ($n = 1$), and a bound $|f^{(n)}| \leq M$ turns it into the error estimate $|f(\beta) - P(\beta)| \leq \frac{M}{n!}|\beta - \alpha|^n$

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Key takeaways from this section:

- **Theorem 5.15 (Taylor):** $f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$ for some x between α and β , where $P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t - \alpha)^k$
- **It generalizes the MVT** ($n = 1$), and a bound $|f^{(n)}| \leq M$ turns it into the error estimate $|f(\beta) - P(\beta)| \leq \frac{M}{n!}|\beta - \alpha|^n$
- **Proof pattern:** auxiliary function $g = f - P - M(t - \alpha)^n$; and a cascade of n mean value theorems

Coming up next: Differentiation of Vector-Valued Functions — what survives (and what fails!) when f takes values in $\mathbb{R}^k$.